

# MENGYU JI

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## EDUCATION

### Ph.D.

*September 2020 – December 2026*

**Zhejiang University** (A joint program with Westlake University)

Major: Computer Science and Technology

Supervisor: Prof. Shiyu Zhao

### M.Eng.

*September 2017 – June 2020*

**University of Chinese Academy of Sciences**

Major: Aerospace Engineering (GPA: 3.61/4)

### B.Eng.

*September 2013 – June 2017*

**Harbin Engineering University**

Major: Aerospace Engineering (GPA: 3.84/4)

## EXCHANGE EXPERIENCE

**Technion – Israel Institute of Technology**

*August 2018, Haifa, Israel*

- Participated in academic seminars on distributed space systems.

**Harbin Institute of Technology (HIT)**

*September 2015 – July 2016, Harbin, China*

- Completed coursework in flight mechanics and control systems.

## RESEARCH INTERESTS

- Robust flight control, physical interaction control for aerial manipulators.
- Learning-based manipulation of unknown ground objects.
- System-level implementation for robotic systems.

## PUBLICATIONS

- 1 **M. Ji**, S. Guo, Z. Li, J. Shen, H. Cao\*, and S. Zhao, PreGME: Prescribed Performance Control of Aerial Manipulators based on Variable-Gain ESO, IEEE Transactions on Industrial Electronics (TIE), 2026. [Paper Link] [Video Link] [Code Link]

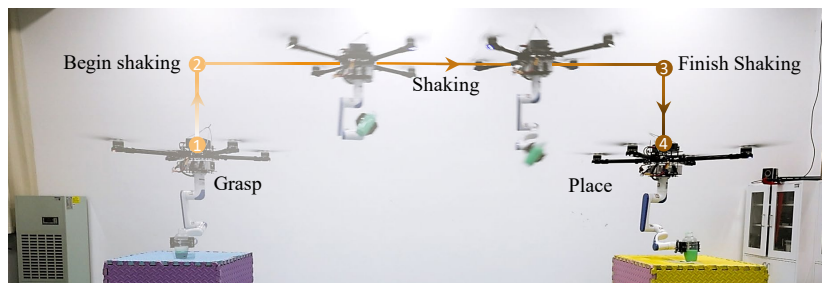


Figure 1: Aerial mixology experiment.

- 2 Z. Li, J. Shen, **M. Ji**, H. Cao\*, and S. Zhao, Predictive Kinematic Coordinate Control for Aerial Manipulators Based on Modified Kinematics Learning, 2025 IEEE International Conference on Robotics and Automation (ICRA), 2025. [Paper Link]
- 3 **M. Ji**, J. Shen, H. Cao\*, and S. Zhao, Avian-Inspired High-Precision Tracking Control for Aerial Manipulators, Non-linear Dynamics, 2024. [Paper Link]

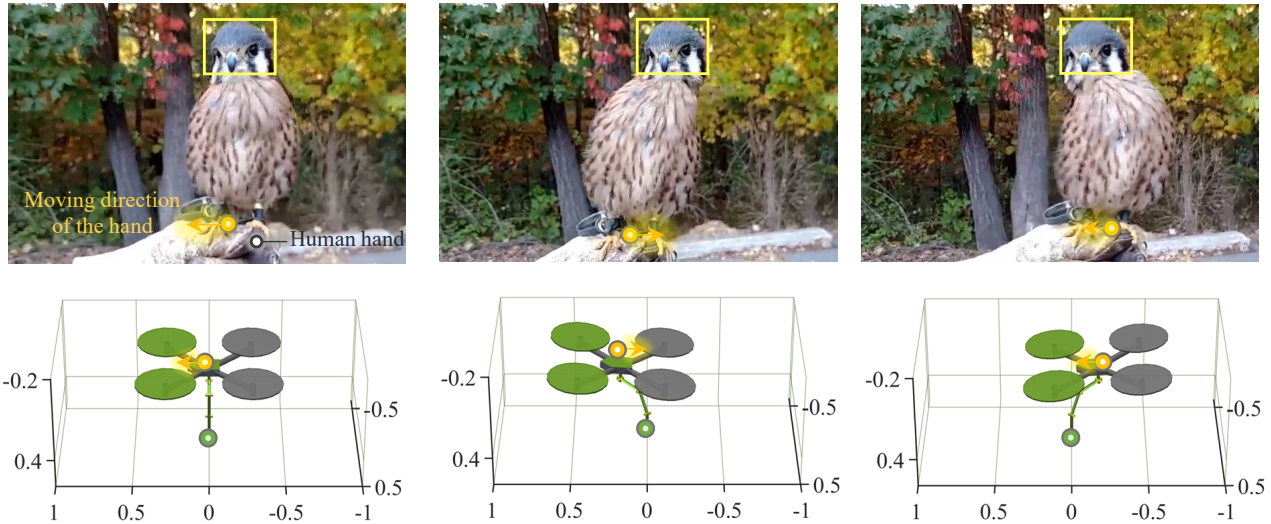


Figure 2: The visual description of the avian-inspired end-effector pose stabilization.

- 4 **M. Ji**, L. Zhang, and S. Wang\*, A Path Planning Approach Based on Q-learning for Robot Arm, 3rd International Conference on Robotics and Automation Sciences (ICRAS), 2019. (**Best Oral Presenter Award**) [Paper Link]

## MANUSCRIPTS IN PREPARATION

- 1 **M. Ji**, J. Li, Z. Li, H. Cao and S. Zhao, Aerial Cart-Pulling: Inverse-Model-Based Interaction Control without Force Sensors, 2026.

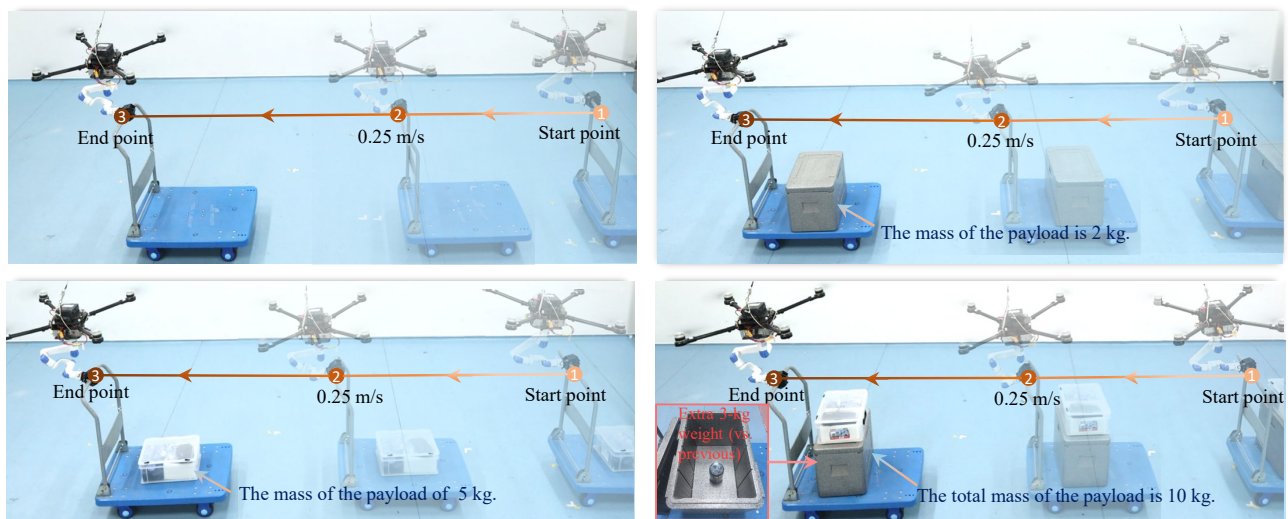


Figure 3: Aerial cart-pulling experiment.

## CONFERENCE PRESENTATIONS

[**Oral Presentation**], Avian-Inspired High-Precision Tracking Control for Aerial Manipulators, 8th IEEE International Conference on Unmanned Systems (ICUS), Changzhou, China, 2025.

[**Oral Presentation**], Avian-Inspired High-Precision Tracking Control for Aerial Manipulators, International Conference on Frontiers of Intelligent Manufacturing and Automation (ICFIMA), Baotou, China, 2024.

## RESEARCH EXPERIENCE

### Aerial Cart-Pulling: Inverse-Model-Based Interaction Control without Force Sensors

2025 – 2026

- Developed a trajectory-based interaction control strategy utilizing meta-learning to generate desired pulling forces for various payloads.
- Eliminated the reliance on real-time force measurements by mapping the desired pulling force directly to the end-effector trajectory via an inverse model.

### Motion Control for Underactuated Aerial Manipulators

2023– 2025

- Developed a partially decoupled motion control framework based on the variable-gain extended state observer (ESO) and error trajectory constraints for underactuated aerial manipulators.
- Built an aerial manipulator hardware system (quadcopter + 6-DOF robotic arm) and validated the framework through three tasks: Aerial Staff Twirling, Aerial Mixology, and Aerial Cart-Pulling.

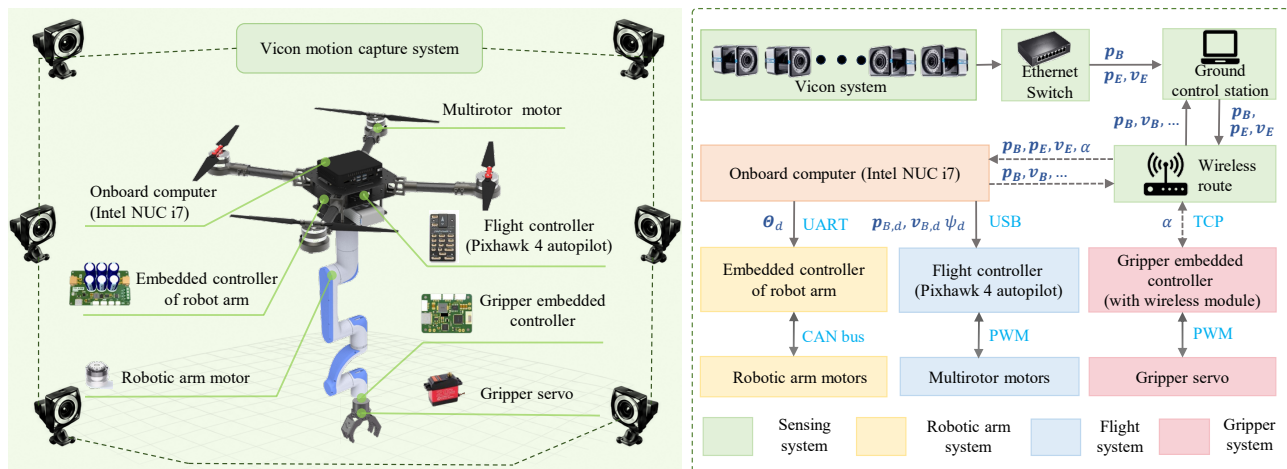


Figure 4: Hardware configuration and system architecture of the aerial manipulator.

### Tracking Control for Underactuated Aerial Manipulators

2020 – 2023

- Designed a partially decoupled motion control method based on the Recursive Newton-Euler (RNE) approach to achieve high-precision tracking for multi-DOF aerial robotic systems.

### Path Planning for Robot Arms

2018 – 2020

- Proposed a path planning method for a 3-DOF fixed-base robotic arm using Q-learning.

## TEACHING EXPERIENCE

### Teaching Assistant, Robot Modeling and Control

2025 – 2026

- Design lab curriculum and develop reference code for simulation tasks.
- Instruct weekly laboratory sessions, guiding students in system modeling and control execution.

## HONORS AND AWARDS

- 1 **Media Coverage:** Research on aerial manipulators was featured by national media, including CCTV-13 (China Central Television) and Bilibili, [Video Link]

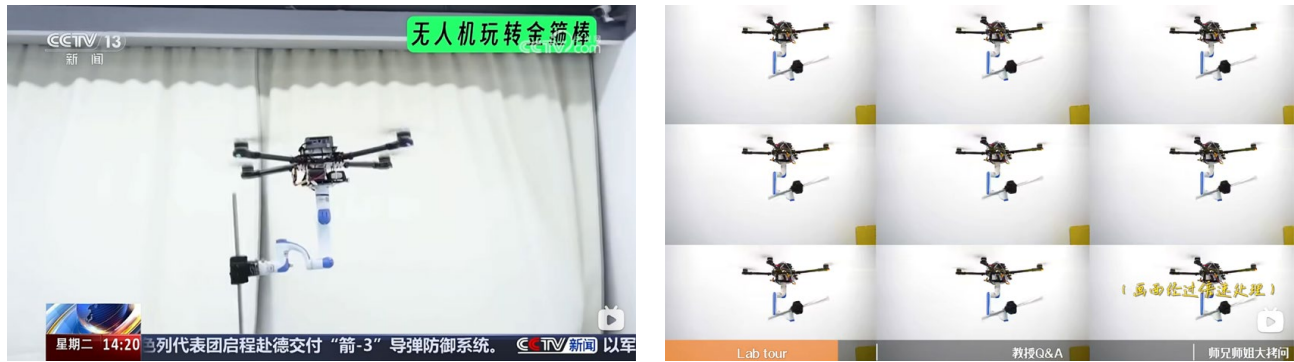


Figure 5: National media coverage (CCTV-13) showcasing the dynamic aerial staff-twirling experiment.

- 2 Best Presentation Award, the Three-Minute Presentation Competition at Westlake University, 2024.
- 3 Best Oral Presenter Award, 3rd International Conference on Robotics and Automation Sciences, 2019.

## RESEARCH SKILLS

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- **Programming Languages:** C/C++, Python, MATLAB/Simulink
- **Robotics & Simulation:** ROS (Robot Operating System), Gazebo
- **Language:** Chinese (Native), English (Fluent)